

Lecture :- Arrays 2

Agenda

find k in row and col wise
sorted matrix

first missing +ve integer.

Sum of all submatrices sum.

Ques Given a row and col wise sorted matrix.

find out whether unique k is present or not. { Microsoft Product }

	0	1	2	3
0	-5	-2	1	13
1	-4	0	3	14
2	-3	2	6	18

k=13 true.

k=2 true

k=15 false.

Brute force:

```
boolean search(int[][] mat, int k) {  
    n = mat.length  
    m = mat[0].length  
    for(i=0; i<n; i++) {  
        for(j=0; j<m; j++) {  
            if(mat[i][j]==k) {  
                return true;  
            }  
        }  
    }  
    return false;  
}
```

TC: $O(n*m)$
SC: $O(1)$

Expectation :

TC: $O(n+m)$. ——— linear.
SC: $O(1)$

Approach 2

starting pt

$k = 13$

	0	1	2	3
0	5	10	15	20
1	6	12	18	24
2	7	14	21	28
3	8	16	24	32

starting pt

false

	0	1	2	3
0	5	10	15	20
1	6	12	18	24
2	7	14	21	28
3	8	16	24	32

$k = 16$
true

	0	1	2	3
0	5	10	15	20
1	6	12	18	24
2	7	14	21	28
3	8	16	24	32

$k = 32$
true

Algorithmic code

```
boolean search(int[][] mat, int k) {  
    n = mat.length  
    m = mat[0].length  
    i = 0  
    j = m - 1 } top right  
    while (i < n && j >= 0) {  
        if (mat[i][j] == k) {  
            return true;  
        }  
        if (mat[i][j] > k) {  
            j--;  
        } else {  
            i++;  
        }  
    }  
    return false;  
}
```

TC: $O(n+m)$

SC: $O(1)$

Submatrices and its identification

Subarray is continuous and ordered part of array

Submatrix is " " " " " matrix.

	0	1	2	3	4
0	1	2	3	4	5 ✓
1	6 ✓	7	8	9 ✓	10
2	11	12	13	14	15
3	16	17	18	19	20 ✓
4	21	22	23	24	25

Submatrices

Block X.

1	2	3	4
---	---	---	---

Subarrays:

1
1 2
1 2 3
1 2 3 4
2
2 3
2 3 4
3
3 4
4

Break: 8:16 - 8:26

Problem 5

Sum of all submatrices sum

[Microsoft
Walmart
Amazon]

Given mat[n][m], determine sum of all the possible submatrices.

	0	1	2
0	4	9	6
1	5	-1	2

All submatrices

$$[4] = 4$$

$$[9] = 9$$

$$[6] = 6$$

$$[5] = 5$$

$$[-1] = -1$$

$$[2] = 2$$

$$[4 \ 9] = 4 + 9$$

$$[9 \ 6] = 9 + 6$$

$$[5 \ -1] = 5 - 1$$

$$[-1 \ 2] = -1 + 2$$

$$\begin{bmatrix} 4 \\ 5 \end{bmatrix} = 4 + 5$$

$$\begin{bmatrix} 9 \\ -1 \end{bmatrix} = 9 - 1$$

$$\begin{bmatrix} 6 \\ 2 \end{bmatrix} = 6 + 2$$

$$\begin{bmatrix} 4 & 9 \\ 5 & -1 \end{bmatrix} = 4 + 9 + 5 - 1$$

$$\begin{bmatrix} 9 & 6 \\ -1 & 2 \end{bmatrix} = 9 - 1 + 6 + 2$$

$$\begin{bmatrix} 4 & 9 & 6 \\ 5 & -1 & 2 \end{bmatrix}$$

$$[4 \ 9 \ 6]$$

$$4 + 9 + 6$$

$$[5 \ -1 \ 2]$$

$$= 5 - 1 + 2$$

$$4 + 9 + 6 + 5 - 1 + 2$$

$$\text{sum} = 4 * 6 + 9 * 8 + 6 * 6 + 5 * 6 + (-1) * 8 + 2 * 6$$

$$= 24 + 72 + 36 + 30 - 8 + 16$$

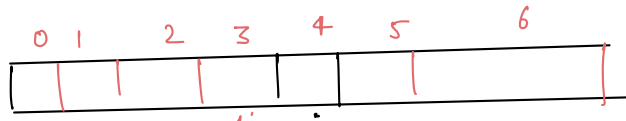
$$= \underline{\underline{\text{Ans}}}$$

Approach

$$4 * 6 + 9 * 8 + 6 * 6 + 5 * 6 + (-1) * 8 + 2 * 6$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $[0][0] \text{ occ} \quad [0][1] \text{ occ} \quad [0][2] \quad \dots$

	0	1	2
0	4	9	6
1	5	-1	2



i is part of how many subarrays?

$$(i+1) * (n-i)$$

$\uparrow$ start indices
 $\uparrow$ end indices

Challenge

$A[10][10]$

	0	1	2	3	4	5	6	7	8	9
0	✓	✓	✓	✓						
1	✓	✓	✓	✓						
2	✓	✓	✓	✓						
3	✓	✓	✓	(i,j)	✓	✓	✓	✓	✓	✓
4				✓	✓	✓	✓	✓	✓	✓
5				✓	✓	✓	✓	✓	✓	✓
6				✓	✓	✓	✓	✓	✓	✓
7				✓	✓	✓	✓	✓	✓	✓

$(j+1)$ (above row 3)
 $(i+1)$ (left of column 3)
 $(n-i)$ (below row 3)
 $(m-j)$ (right of column 3)

No of start indices where $(3,3)$ can be part of

$$\Rightarrow (i+1) * (j+1)$$

No of end indices where $(3,3)$ can be part of

$$\Rightarrow (n-i) * (m-j)$$

$$\text{occ} \Rightarrow \left\{ \frac{(i+1) * (j+1)}{\text{top left start}} * \frac{(n-i) * (m-j)}{\text{bottom right end}} \right\}$$

int totalSubMatrixSum(int[][] mat) {

total = 0

for (i = 0; i < n; i++) {

for (j = 0; j < m; j++) {

start = (i+1) * (j+1)

end = (n-i) * (m-j)

occ = start * end;

total = total +

A[i][j] * occ;

}

return total;

}

	0	1	2	3
0	5	10	15	20
1	6	12	18	24
2	7	14	21	28
3	8	16	24	32

14 → 2, 1

$6 * 6 = 36$
 $(i+1)(j+1) * (n-i)(m-j)$

$(3 * 2) * (2 * 3)$

$6 * 6 = 36$ Ans

TC: $O(n * m)$

SC: $O(1)$

Break: 10: 18: 10: 28

Qn Given arr[n], find first missing +ve number. Google

3	-2	1	2	7
---	----	---	---	---

ans = 4

1	2	5	6	4	3
---	---	---	---	---	---

ans = 7

1	0	-5	-6	4	2
---	---	----	----	---	---

ans = 3

$$5 \quad 3 \quad 1 \quad -1 \quad -2 \quad -4 \quad 7 \quad 2 \longrightarrow \text{Ans} = 4 \text{ Ans}$$

$-1 \quad -3 \quad -2 \quad -4 \longrightarrow \text{Ans} = 1$

Claim $\text{Ans} = [1, n+1]$

$\uparrow$
len

Approach 1 Brute force approach.

its form 1 to $n+1$ —

search in array

```

if not found
    return on
else
    continue

```

TC: $O(n^2)$
SC: $O(1)$

sorting, hashmap $O(n^2)$ $O(n^2)$ } $\rightarrow$ future classes

TC: $O(n^2)$

SC: 001' $\longrightarrow$ cannot create new space

Use same array

A | p | p | r | u | o | a | c | h | 2

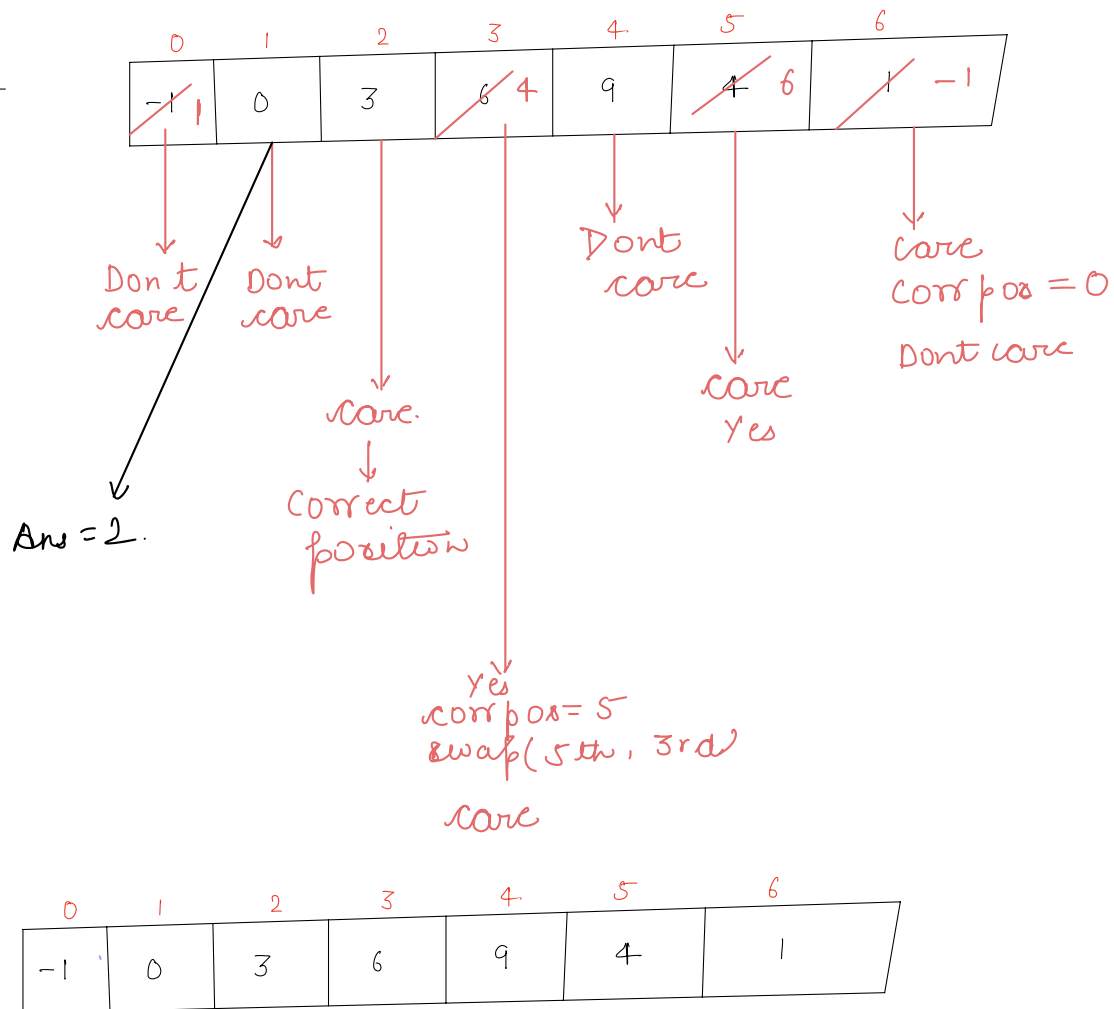
Expected solution

Intuition

0	1	2	3	4	5	6
1	2	3	4	5	6	7

Ans = [1, 8]

Example:



0	01	1	2	3	4	5
2 4	4 2	1 2	2 4	-10	7	

1. care
curr pos = 1
swap(0th, 1st)

care.
No swap

care
No swap

Dont care

Dont care

care
curr pos = 0
swap(0, 2)

2. care
curr pos = 3
swap(0th, 3rd)

care.
curr pos = 1.

3. care
curr pos = 1.
swap(0th, 1st)

can give you infinite loop
2 is already present at its correct id.
stop

0	1	2	3	4	5
2	4	1	2	-10	7

Algorithmic code

```
int firstMissingInterval(int[] arr) {  
    i = 0  
    while (i < n) {  
        if (A[i] >= 1 && A[i] <= n) {  
            correctIdx = A[i] - 1;  
            if (A[i] != A[correctIdx]) {  
                swap(A, i, correctIdx);  
            } else {  
                i++;  
            }  
        } else {  
            i++;  
        }  
    }  
    for (i = 0; i < n; i++) {  
        if (A[i] != i + 1) {  
            or  
            A[i] - 1 != i  
            return i + 1;  
        }  
    }  
    return n + 1;  
}
```

$O(n)$

$O(n)$

TC: $O(n)$ ✓
S: $O(1)$

Thankyou 😊

Doubt

0	1	2	3
5	6	7	8

↑

0	1	2	3	4	5
2	4	1	2	-10	7